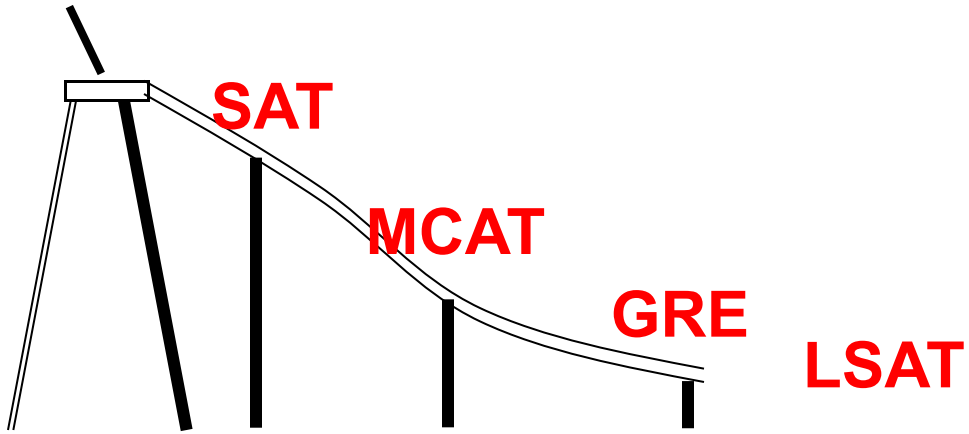
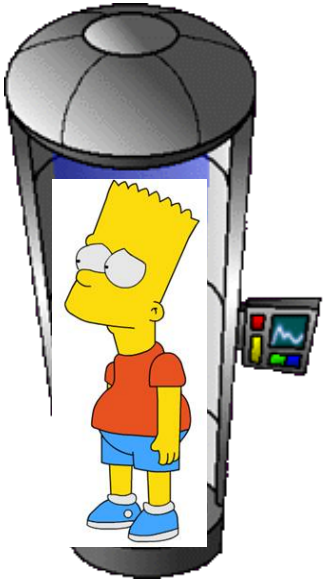
Test Slide



Quantum Communication





Dave Bacon

Department of Physics

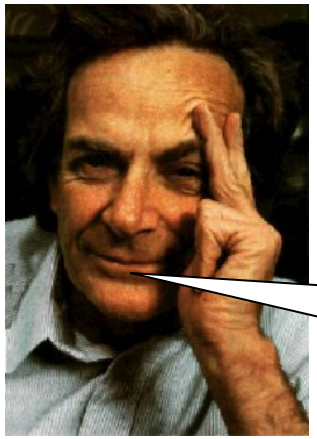
Institute for Quantum Information

Caltech





Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922



They understand it.
They use it.
But they don’t really believe it?

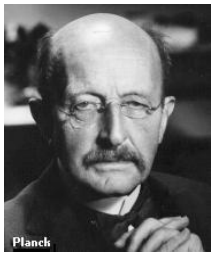
Quantum History

Inventin

Worrying

Acceptin g

1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000



Max Planck



Niels Bohr



Erwin Schrodinger



John Von Neumann



John Bell



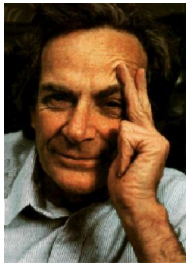
Peter Shor



Louis de Broglie



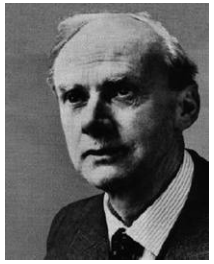
Werner Heisenberg



Richard Feynman



John Archibald Wheeler



P.A.M. Dirac

Acceptance

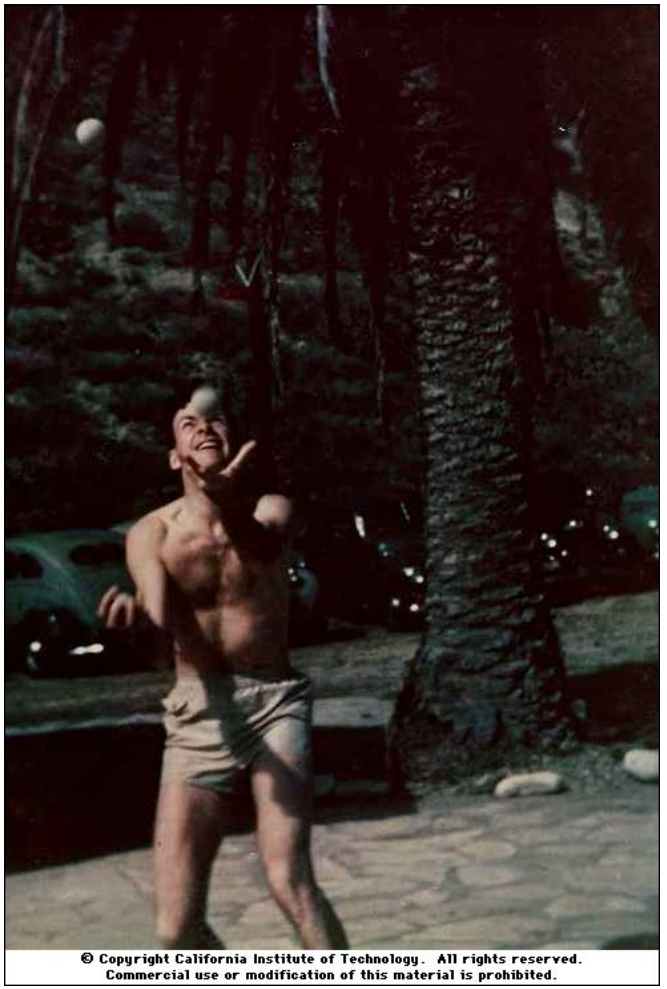
on quantum theory:

"Do not ask yourself, if you can possibly avoid that, 'how can it be like that?' because you will lead yourself down a blind alley in which no one has ever escaped."

"For a successful technology, nature must take precedence over public relations, for nature cannot be fooled."

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer?

It's not a Turing machine but a machine of a different kind" (1981)

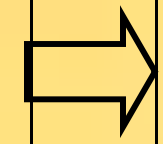

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Richard Feynman

Quantum Radio

probability	
0	p_0
1	$1 - p_0$

bit
s

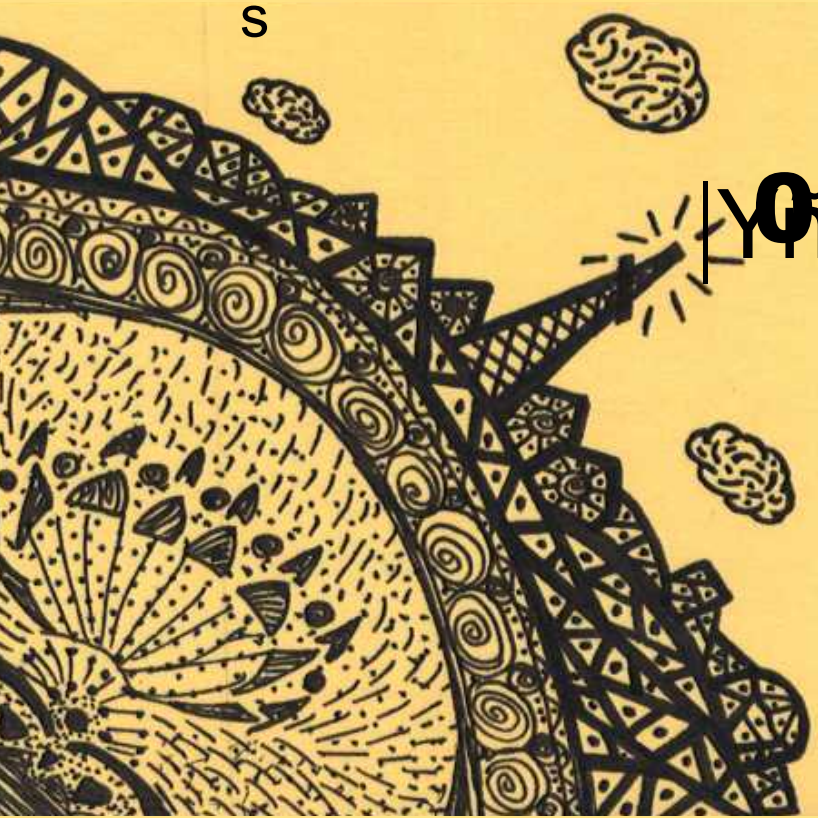


amplitude
s

$$|\psi\rangle \triangleq \alpha|0\rangle + \beta|1\rangle$$

qubits



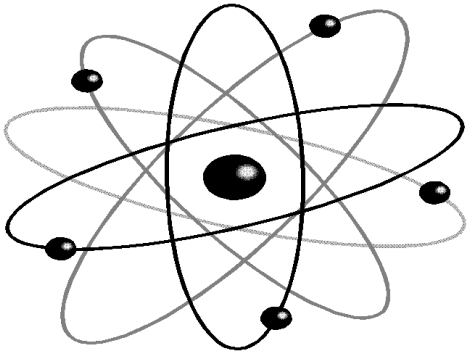




01001011011

4 Minute Quantum Mechanic

Postulate 1: Associated with any isolated physical system is a complex vector space with inner product known as the *state space* of the system. The system is completely described by its *state vector*, which is a unit vector in the system's state space.



physical system

complex vector space

What's a Qubit?

Classical State Descriptions



0



1

Suppose we don't have information about the bit classically. How do we describe the state?

p probability of system being in state 0

$1-p$ probability of system being in state 1

vector of probabilities: $\begin{bmatrix} p \\ 1-p \end{bmatrix}$

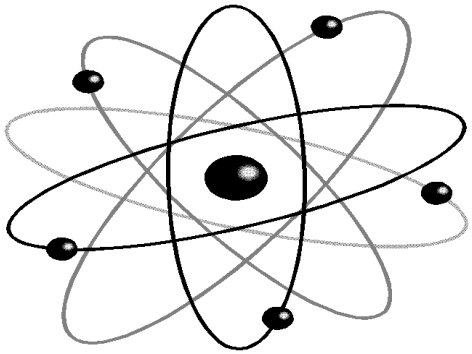
Quantum State Descriptions

vector of amplitudes: $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ complex numbers

$|\alpha|^2 + |\beta|^2 = 1$

$\alpha|0\rangle + \beta|1\rangle$

a qubit

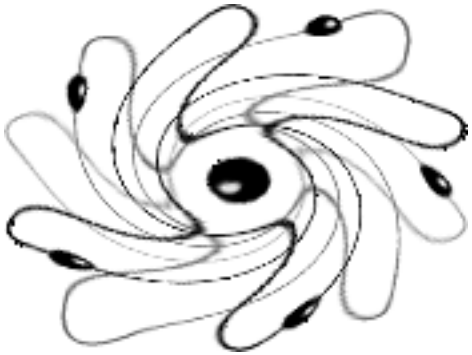


physical system

complex vector space

complex numbers

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{d-1} \end{bmatrix}$$



state of system

complex unit vector

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{d-1} \end{bmatrix}$$

$$\alpha_0|0\rangle + \alpha_1|1\rangle + \cdots + \alpha_{d-1}|d-1\rangle$$

$$|\alpha_0|^2 + |\alpha_1|^2 + \cdots + |\alpha_{d-1}|^2 = 1$$

What's a Qubit?

$$|0\rangle \qquad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle \qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\tfrac{1}{2}|0\rangle + i\tfrac{\sqrt{3}}{2}|1\rangle \qquad \begin{bmatrix} \frac{1}{2} \\ i\frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\alpha|0\rangle + \beta|1\rangle \qquad \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

3 Minute Quantum Mechanic

Postulate 2: The evolution of a closed quantum system is described by a *unitary transformation*. That is the state $|\psi\rangle$ of the system at time t1 is related to the state $|\psi'\rangle$ of the system at time t2 by a unitary $\mathbf{U}$ operator which depends only on the times t1 and t2.

$$|\psi'\rangle = \mathbf{U}|\psi\rangle \qquad \mathbf{U}\mathbf{U}^\dagger = \mathbf{I}$$

Classical State Evolution

If the state is 0 flip to 1 with probability s (to 0 with probability 1-s)
If the state is 1 flip to 1 with probability t (to 0 with probability 1-t)

$$\begin{bmatrix} p \\ 1-p \end{bmatrix} \longrightarrow \begin{bmatrix} p' \\ 1-p' \end{bmatrix}$$

$$\begin{bmatrix} p' \\ 1-p' \end{bmatrix} = \begin{bmatrix} 1-s & 1-t \\ s & t \end{bmatrix} \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

Quantum State Evolution

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \longrightarrow \begin{bmatrix} \alpha' \\ \beta' \end{bmatrix}$$

$$\begin{bmatrix} \alpha' \\ \beta' \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$|\psi'\rangle$ $\mathbf{U}$ $|\psi\rangle$

unitarity $\mathbf{U}\mathbf{U}^\dagger = \mathbf{I}$

$$\begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} u_{00}^* & u_{10}^* \\ u_{01}^* & u_{11}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3 Minute Quantum Mechanic

Postulate 2: The evolution of a closed quantum system is described by a *unitary transformation*. That is the state $|\psi\rangle$ of the system at time t1 is related to the state $|\psi'\rangle$ of the system at time t2 by a unitary operator $\mathbf{U}$ which depends only on the times t1 and t2,

$$|\psi'\rangle = \mathbf{U}|\psi\rangle \qquad \mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$$

$$\begin{bmatrix} \alpha'_0 \\ \alpha'_1 \\ \vdots \\ \alpha'_{d-1} \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} & \cdots & u_{0d} \\ u_{10} & u_{11} & & \vdots \\ \vdots & & \ddots & \vdots \\ u_{d0} & \cdots & \cdots & u_{dd} \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{d-1} \end{bmatrix}$$

Some Quantum Evolution

$$\mathbf{U} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\mathbf{U}|0\rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$$

$$\mathbf{U}|1\rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle$$

$$\mathbf{U}\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}\right) = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}+1}{2\sqrt{2}} \\ \frac{1-\sqrt{3}}{2\sqrt{2}} \end{bmatrix}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}|0\rangle + \frac{1-\sqrt{3}}{2\sqrt{2}}|1\rangle$$

2 Minute Quantum Mechanic

Postulate 3: Quantum measurements are described by a collection $\{\mathbf{M}_m\}$ of *measurement operators.* These are operators acting on the state space of the system being measured. The index m refers to The measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement, then the probability of result m is given by

$$p_m = \langle\psi|\mathbf{M}_m^\dagger \mathbf{M}_m|\psi\rangle$$

and the state of the system after the measurement is

$$\frac{\mathbf{M}_m|\psi\rangle}{\sqrt{p_m}}$$

The measurement operators satisfy

$$\sum_m \mathbf{M}_m^\dagger \mathbf{M}_m = \mathbf{I}$$

Classical State Measurement

$$\begin{bmatrix} p \\ 1-p \end{bmatrix}$$



probability p we get heads $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

probability 1-p we get tails $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Quantum State Measurement

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

probability $|\alpha|^2$ we get heads outcome 0 $\begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$

probability $|\beta|^2$ we get heads outcome 1 $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$

1 Minute Quantum Mechanic

Postulate 4: The state space of a composite physical system is the Tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and system number i is prepared in the state ρ_i, then the joint state of the total system is

$$\rho_1 \otimes \rho_2 \otimes \cdots \rho_n$$

Two Classical Systems

Coin A

Coin B



00



01



10



11

$$\begin{bmatrix} p_{00} \\ p_{01} \\ p_{10} \\ p_{11} \end{bmatrix}$$

If we prepare Coin A in

$$\begin{bmatrix} p \\ 1-p \end{bmatrix}$$

and if we prepare Coin B in

$$\begin{bmatrix} q \\ 1-q \end{bmatrix}$$

then

$$\begin{bmatrix} p_{00} \\ p_{01} \\ p_{10} \\ p_{11} \end{bmatrix} = \begin{bmatrix} pq \\ p(1-q) \\ (1-p)q \\ (1-p)(1-q) \end{bmatrix}$$

Two Quantum Systems

qubit A qubit B

$$|0\rangle \otimes |0\rangle = |0\rangle|0\rangle = |00\rangle$$

$$|0\rangle \otimes |1\rangle = |0\rangle|1\rangle = |01\rangle$$

$$|1\rangle \otimes |0\rangle = |1\rangle|0\rangle = |10\rangle$$

$$|1\rangle \otimes |1\rangle = |1\rangle|1\rangle = |11\rangle$$

$$\begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix} = \alpha_{00}|0\rangle \otimes 0\rangle + \alpha_{01}|0\rangle \otimes |1\rangle + \alpha_{10}|1\rangle \otimes |0\rangle + \alpha_{11}|1\rangle \otimes |1\rangle$$

qubit A as $\alpha|0\rangle + \beta|1\rangle$

qubit B as $\gamma|0\rangle + \delta|1\rangle$

$$\Rightarrow \quad (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) = \begin{bmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{bmatrix}$$

Quantum Radio

probability	
0	p_0
1	$1 - p_0$

bit
s

$$|\psi\rangle \triangleq \alpha|0\rangle + \beta|1\rangle$$

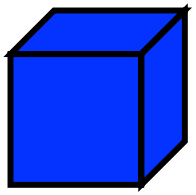
amplitude
s

qubits



From Here to There







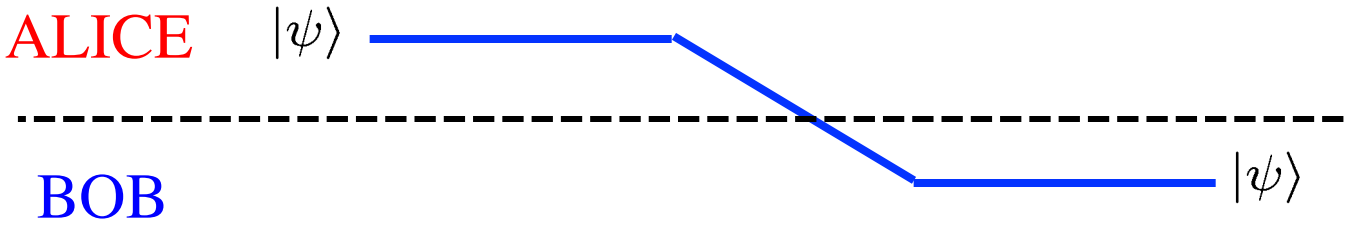
ALICE

some vast distance

(The shortest distance between two points is under construction.)

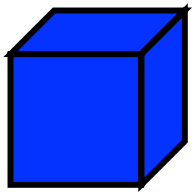
BOB

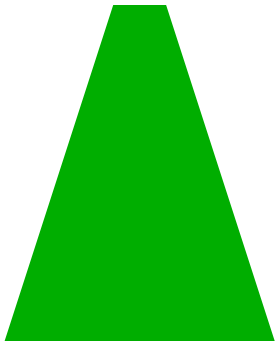
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



From Here to There









ALICE

some vast distance

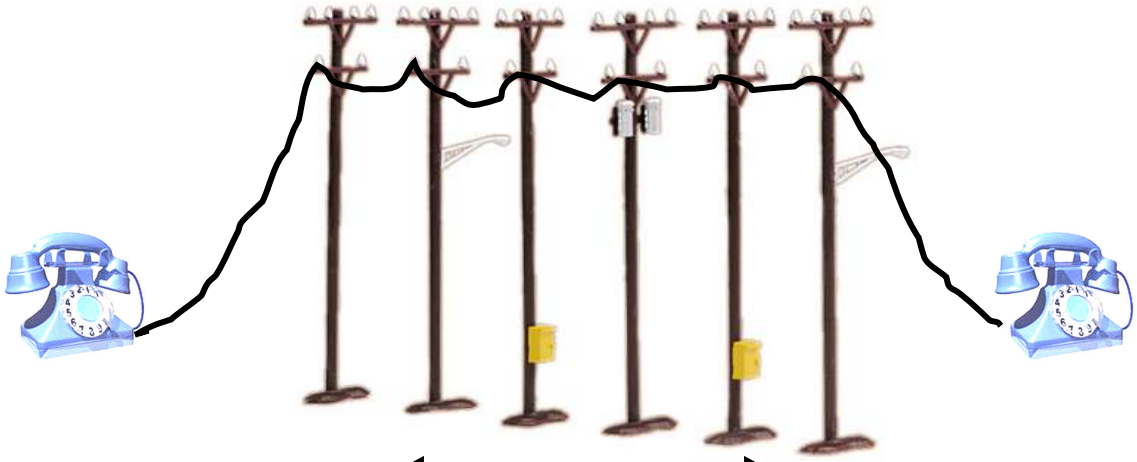
(The shortest distance between two points is under construction.)

BOB

Question: Other methods to transport quantum information?

Silly Question, Silly Answer







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits



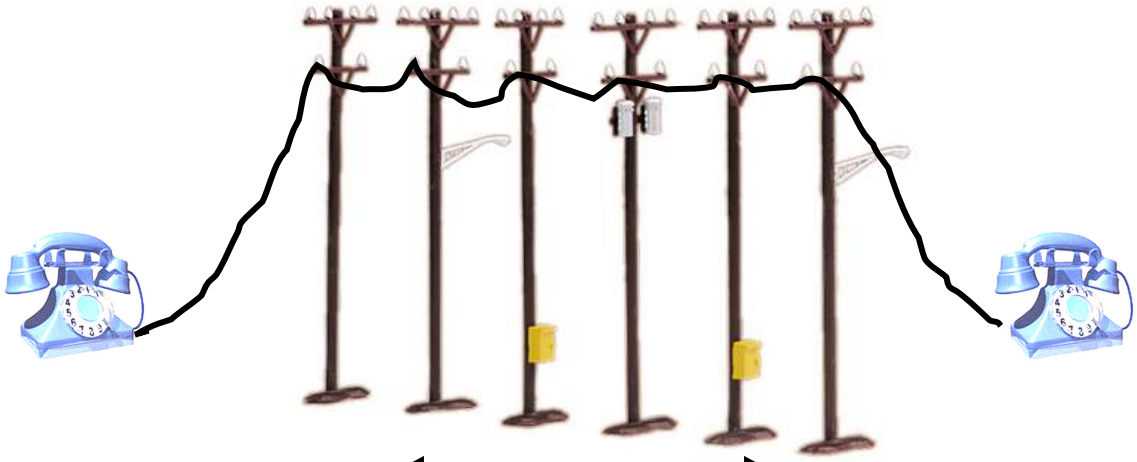
two problems:

infinite number of bits to specify $|\psi\rangle$

cannot perfectly learn unknown $|\psi\rangle$

Silly Question?







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits

+

shared entanglement

A Silly Question?

Entanglement



ALICE

$$|\psi_A\rangle = \alpha_A|0\rangle + \beta_B|1\rangle \qquad |\psi_B\rangle = \alpha_B|0\rangle + \beta_B|1\rangle$$



BOB

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

separable

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

entangled

Entanglement Is Like…

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$



ALICE

measure in $|0\rangle$, $|1\rangle$ basis

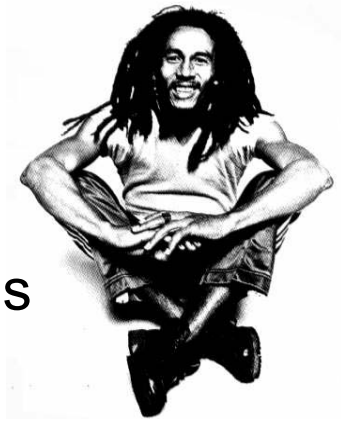
<u>outcome</u>

50% $|0\rangle \otimes |0\rangle$

50% $|1\rangle \otimes |1\rangle$

random correlated bits

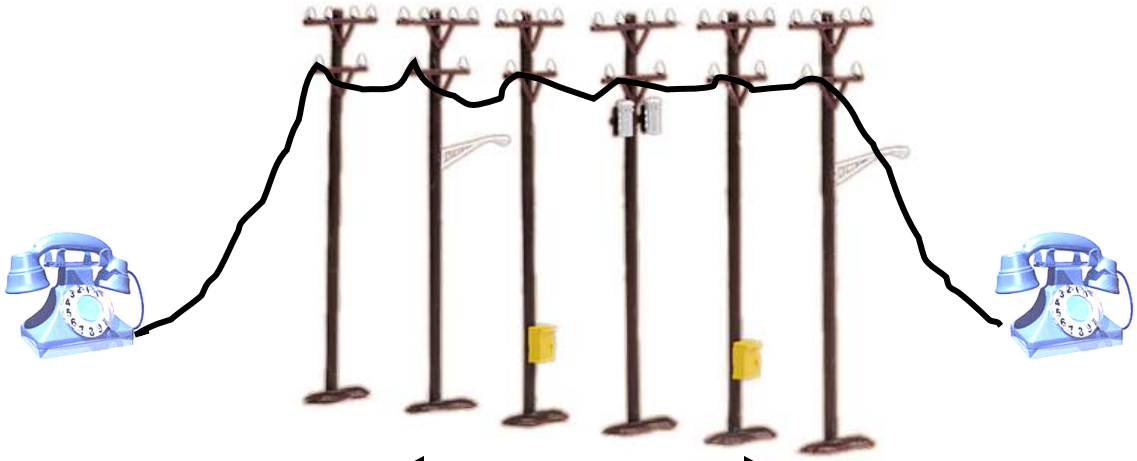
measure in $|0\rangle$, $|1\rangle$ basis



BOB

Silly Question?







ALICE

BOB

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

classical bits

+

random correlated bits

classical bits

random correlated bits

A Silly Question!

Entanglement Is Like…

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$



ALICE

measure in $|0\rangle$, $|1\rangle$ basis

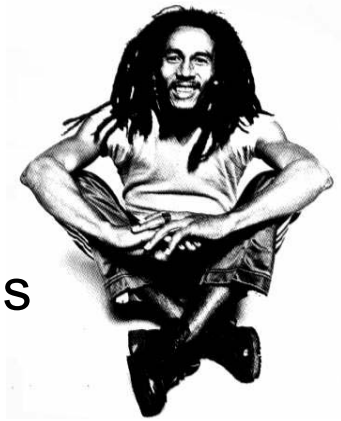
outcome

50% $|0\rangle \otimes |0\rangle$

50% $|1\rangle \otimes |1\rangle$

random correlated bits

measure in $|0\rangle$, $|1\rangle$ basis



BOB

Is It Like Random Correlated Bits?

Bell's Question

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

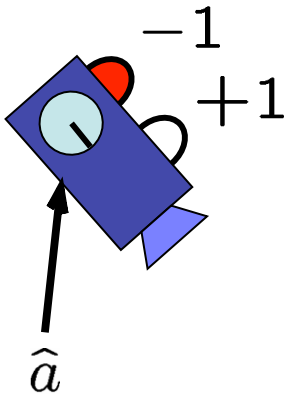


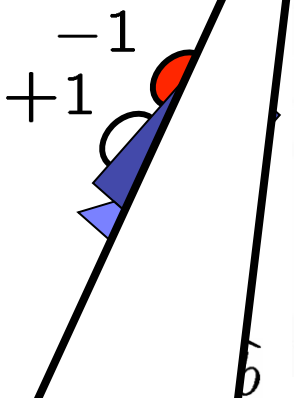
John Bell

A

B









ALICE

$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$

BOB

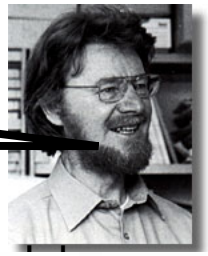
If $\hat{a} = \hat{b}$ then $\langle AB\rangle = -1$, i.e. outputs are "anti"-correlated

$A =$ $= -1$

A $= +1$

Can these correlations be explained by random correlated bits?

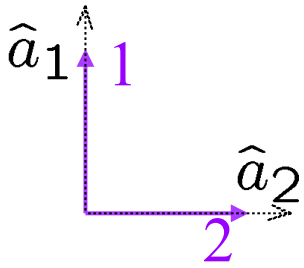
Bell Thinks



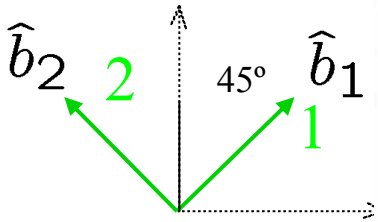
$$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$$



ALICE



	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$



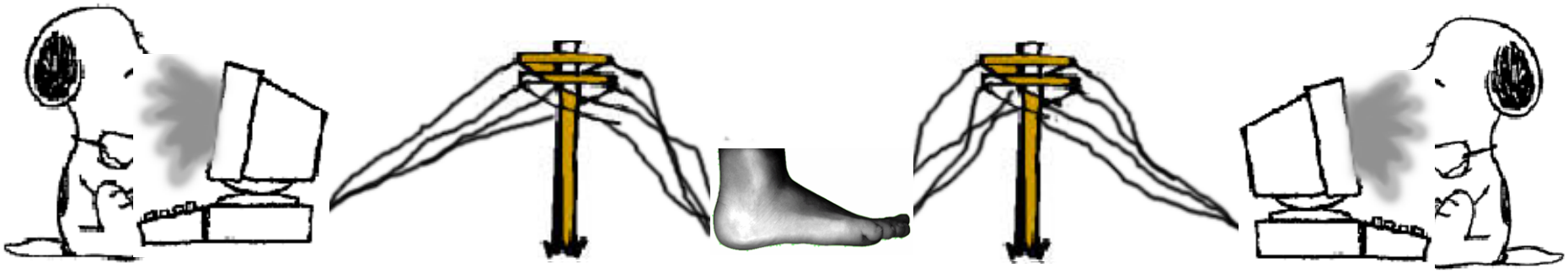


BOB

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

Quantum: $-\frac{4}{\sqrt{2}} = -2\sqrt{2}$

Communication Breakdown



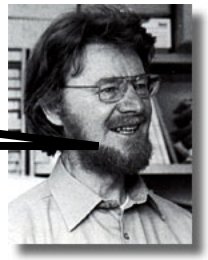
1. Initial unlimited classical communication between computers.
2. Communication lines cuts. **random correlated bits**
3. Input into each computer.
4. Output by each computer.

Repeat

Build up table of input-output correlations

What sort of correlations can be produced?

Bell's Thinks II



"Convex"–iftying a probabilistic protocol

random variables could all be generated in first step of protocol

probabilistic protocol = deterministic protocol with certain probabilities

$$A(\hat{a}_i) \in \{+1, -1\} \qquad B(\hat{b}_i) \in \{+1, -1\}$$

Alice's output and measurement choice
cannot depend on
Bob's output and measurement choice (and vice versa)

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

deterministic protocol

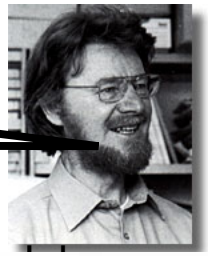
$$A(\hat{a}_1)B(\hat{b}_1) + A(\hat{a}_2)B(\hat{b}_1) + A(\hat{a}_1)B(\hat{b}_2) - A(\hat{a}_2)B(\hat{b}_2)$$

$$A(\hat{a}_1)[B(\hat{b}_1) + B(\hat{b}_2)] + A(\hat{a}_2)[B(\hat{b}_1) - B(\hat{b}_2)] = \pm 2$$

average over shared random bits

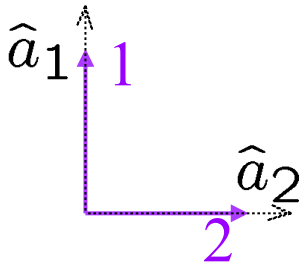
$$|A(\hat{a}_1)[B(\hat{b}_1) + B(\hat{b}_2)] + A(\hat{a}_2)[B(\hat{b}_1) - B(\hat{b}_2)]| \leq 2$$

Bell Answers

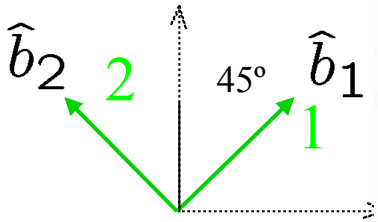


$$\langle A(\hat{a})B(\hat{b})\rangle = -\hat{a}\cdot\hat{b}$$





	1	2
1	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$





ALICE

BOB

$$\langle A(\hat{a}_1)B(\hat{b}_1)\rangle + \langle A(\hat{a}_2)B(\hat{b}_1)\rangle + \langle A(\hat{a}_1)B(\hat{b}_2)\rangle - \langle A(\hat{a}_2)B(\hat{b}_2)\rangle$$

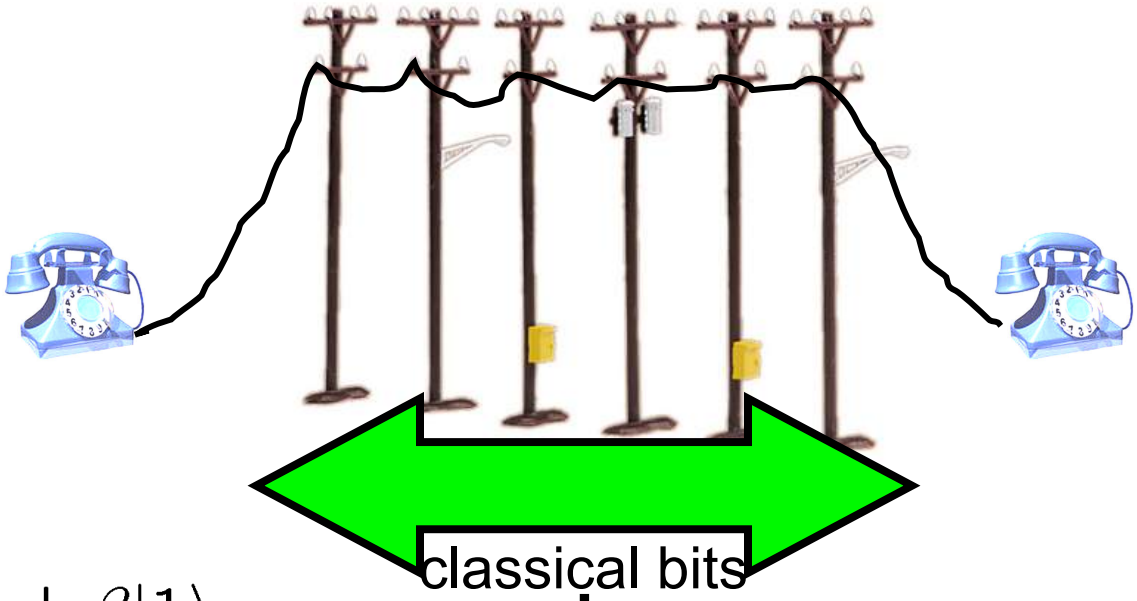
Quantum: $-\frac{4}{\sqrt{2}} = -2\sqrt{2}$

With correlated random bits: between -2 and $+2$

Quantum correlations are DIFFERENT from correlated random bits!

From Here to There







ALICE

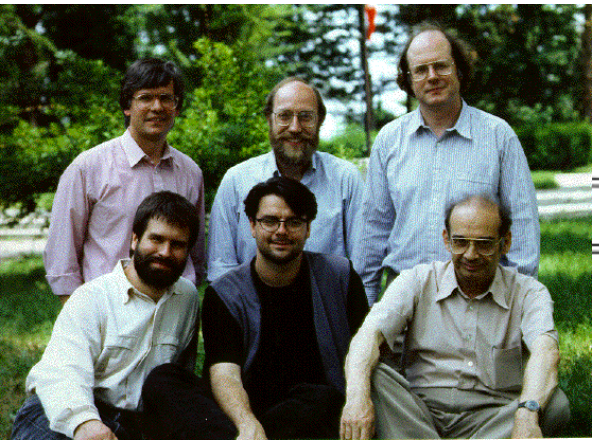
BOB

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

classical bits

+

shared entanglement



PHYSICAL REVIEW
LETTERS

VOLUME 70 29 MARCH 1993 NUMBER 13

Teleporting an Unknown Quantum State via Dual Classical and Einstein-Podolsky-Rosen Channels

Charles H. Bennett,[1] Gilles Brassard,[2] Claude Crépeau,[2,3]
Richard Jozsa,[2] Asher Peres,[4] and William K. Wootters[5]

Bell Pairs

computational basis

$$|0\rangle \otimes |0\rangle$$
$$|0\rangle \otimes |1\rangle$$
$$|1\rangle \otimes |0\rangle$$
$$|1\rangle \otimes |1\rangle$$

Bell basis

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle)$$
$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$
$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$
$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$$

converting

$$|0\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$
$$|0\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$
$$|1\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle - |\Psi_-\rangle)$$
$$|1\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

Bell

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$|\psi\rangle \otimes |\Psi_-\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

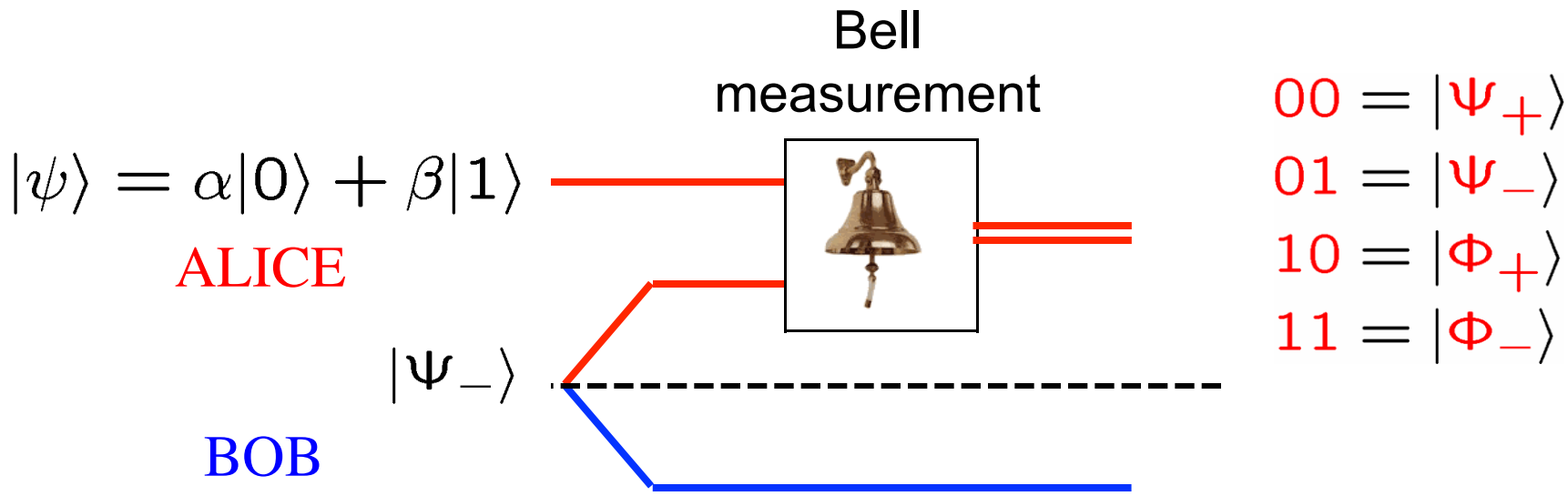
$$= \frac{1}{2}\Big[|\Phi_+\rangle \otimes (\alpha|1\rangle - \beta|0\rangle) + |\Phi_-\rangle \otimes (\alpha|1\rangle + \beta|0\rangle)$$
$$+ |\Psi_+\rangle \otimes (-\alpha|0\rangle + \beta|1\rangle) + |\Psi_-\rangle \otimes (-\alpha|0\rangle - \beta|1\rangle)\Big]$$

$$= \tfrac{1}{2}\Big[|\Phi_+\rangle \otimes \mathbf{XZ}|\psi\rangle + |\Phi_-\rangle \otimes \mathbf{X}|\psi\rangle - |\Psi_+\rangle \otimes \mathbf{Z}|\psi\rangle - |\Psi_-\rangle \otimes |\psi\rangle\Big]$$

$$\mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|\psi\rangle \otimes |\Psi_-\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$= \frac{1}{2}\left[|\Phi_+\rangle \otimes \mathbf{XZ}|\psi\rangle + |\Phi_-\rangle \otimes \mathbf{X}|\psi\rangle - |\Psi_+\rangle \otimes \mathbf{Z}|\psi\rangle - |\Psi_-\rangle \otimes |\psi\rangle\right]$$



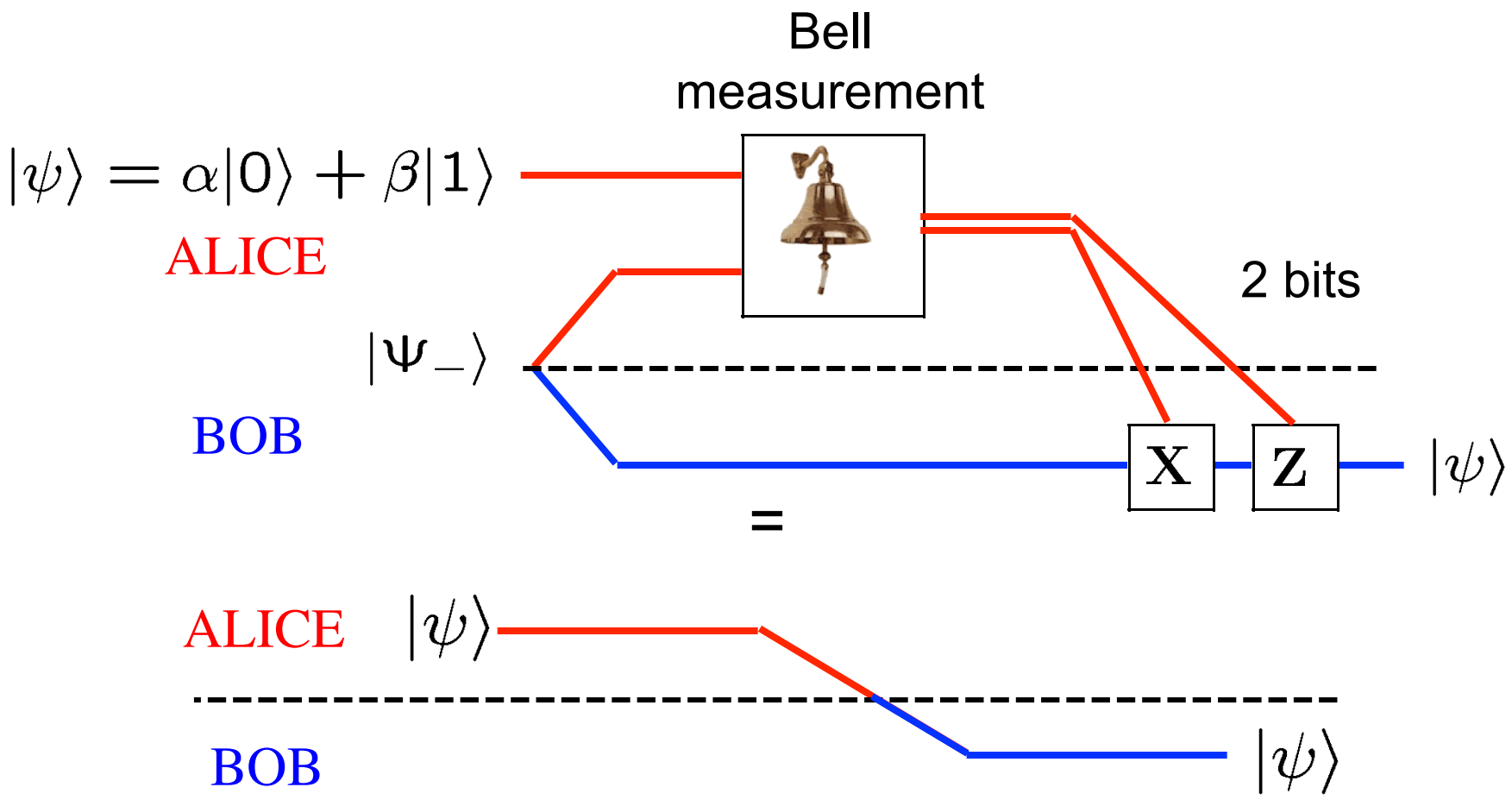
If Alice gets outcome $00 = |\Psi_+\rangle$ then Bob has $\mathbf{XZ}|\psi\rangle$

$\Longrightarrow$ Bob can apply $\mathbf{ZX}$ to $\mathbf{XZ}|\psi\rangle$ and thus Bob now has $|\psi\rangle$

Alice can send her measurement result (2 classical bits) and then Bob can perform the appropriate action $\{\mathbf{ZX}, \mathbf{Z}, \mathbf{X}, \mathbf{I}\}$ to obtain $|\psi\rangle$

Quantum Teleportation!

Quantum Teleportation



1 qubit ≤ 2 classical bits + 1 shared Bell pair

1 qubit ≤ 2 bits + 1 ebit

Quantum Communication

1. Given a *physical resource* (energy, time, bits, ebits, qubits, etc.)

2. Given an *information processing task* (information transmission, data compression, etc.)

3. Given a *success criterion* (exact, approximate, etc.)

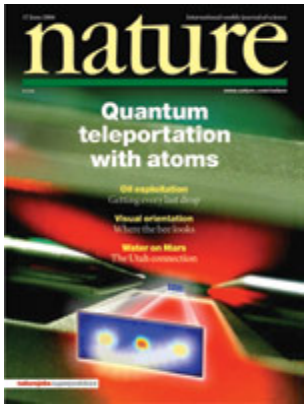
How much of **2.** do we need to achieve **1.** while satisfying **3.**?

(a.k.a. how to write a quant-ph paper!)

Quantum Teleportation

1. ebits and bit
2. transmit a qubit
3. exactly teleport the qubit

1 qubit ≤ 2 bits + 1 ebit

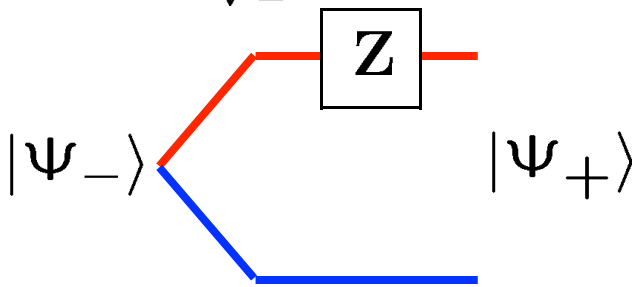


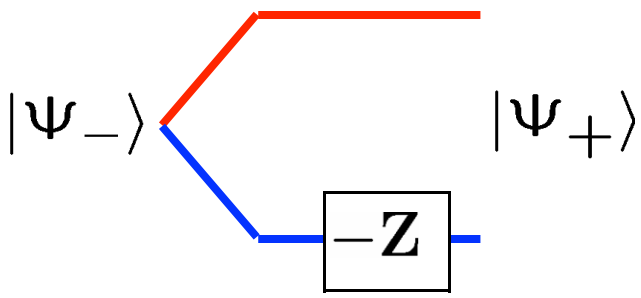
June 16, 2004

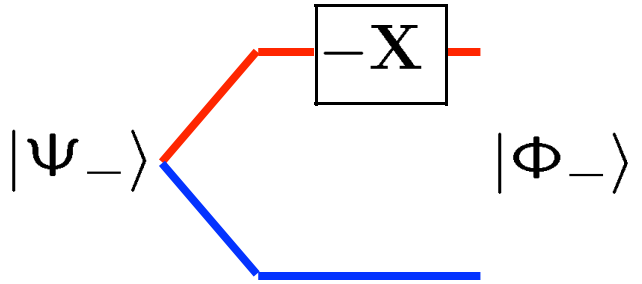
Those

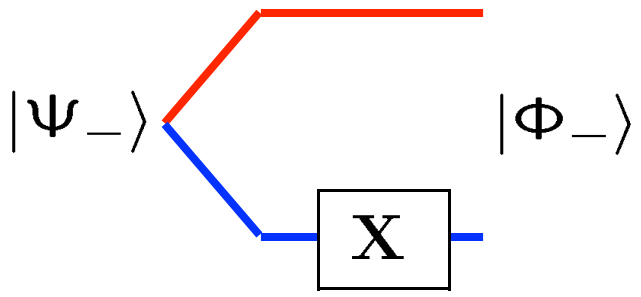
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle) \quad |\Phi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

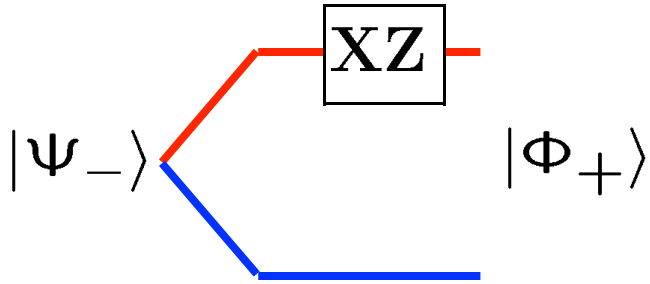
$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle) \quad |\Phi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$$

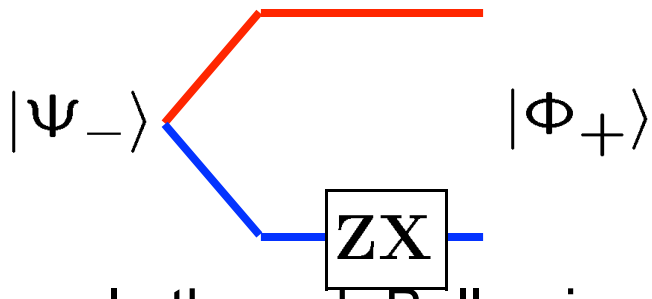












$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Each side can cycle through Bell pairs.

Cycling on Four States



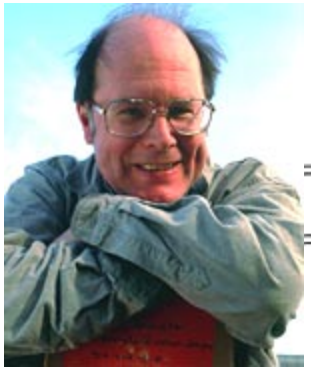
ALICE

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

Alice can cycle through 4 global states



BOB



Charles Bennett

PHYSICAL REVIEW LETTERS

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Communication via One- and Two-Particle Operators on Einstein-Podolsky-Rosen States

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(Received 16 June 1992)



Stephen Wiesner

Superdense Coding



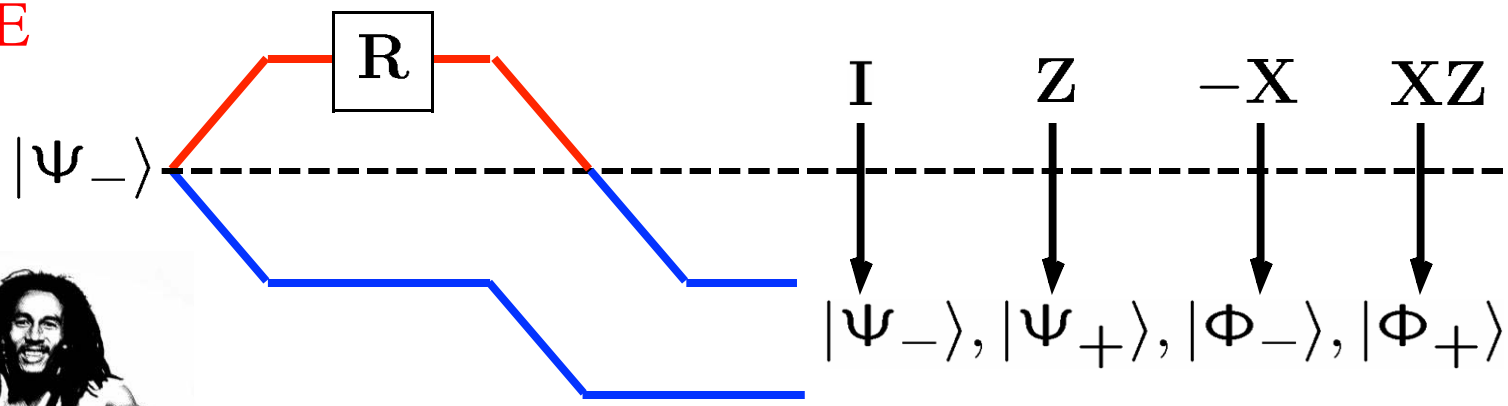
ALICE

1. Alice and Bob share $|\Psi_-\rangle$

2. Alice applies $\mathbf{R}$ to her half of Bell pair

00 01 10 11

$$\mathbf{R} \in \{\mathbf{I}, \mathbf{Z}, -\mathbf{X}, \mathbf{XZ}\}$$





BOB

3. Alice sends her half of Bell pair to Bob

4. Bob measures in Bell basis

00 01 10 11

Superdense Coding

1. Given a *physical resource* (energy, time, bits, ebits, qubits, etc.)

2. Given an *information processing task* (information transmission, data compression, etc.)

3. Given a *success criterion* (exact, approximate, etc.)

1. ebits and qubits
2. transmit a bit
3. exactly communicate bits

2 bits ≤ 1 ebit +1 qubit

Sharing ebits allows us to send 2 bits using 1 qubit

compare teleportation: 1 qubit ≤ 2 bits + 1 ebit

Flavor

1 ebit: $|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$

Quantum information tasks **consume ebits**, ebits are a **resource**

(ebits are a “nonlocal resource”, i.e. shared by localized parties)



Entanglement comes in flavors besides ebits
For example:

$$|\Psi\rangle = \tfrac{1}{2}|0\rangle \otimes |0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle \otimes |1\rangle)$$



How do other flavors of entanglement compare to ebits?

Some Entanglement Is Created Equal

All Bell pairs are equal

$$|\Psi_+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle) \quad \leftrightarrow \quad |\Phi_+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

$$|\Psi_-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle) \quad \leftrightarrow \quad |\Phi_-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$$

When we allow local unitary evolution
(operations Alice and Bob perform separately)

States reachable from

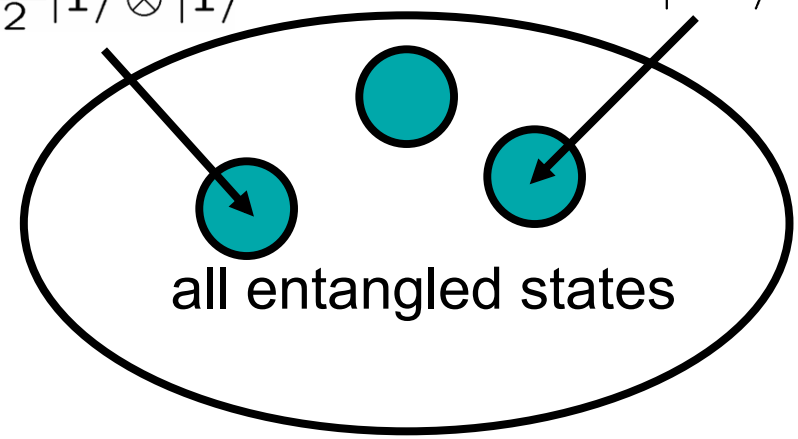
$$|\Psi\rangle = \tfrac{1}{2}|0\rangle \otimes |0\rangle + \tfrac{\sqrt{3}}{2}|1\rangle \otimes |1\rangle$$

by local unitaries

States reachable from

$$|\Psi_-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

by local unitaries



Standardizatio



n generic entangled state

$$|\Psi\rangle = \sum_{i=1}^{d_A} \sum_{j=1}^{d_B} c_{ij} |i\rangle \otimes |j\rangle$$



dimension d_A

dimension d_B

ALICE

BOB

local unitary evolution

$\mathbf{U}_A$

$\mathbf{U}_B$

"Schmidt decomposition"

$$|\Psi\rangle = \sum_{i=1}^{\min(d_A,d_B)} \sqrt{p_i} |i\rangle \otimes |i\rangle$$

$p_i \geq 0, \ \sum_i p_i = 1$

(pure state) two party entanglement classified by Schmidt coefficients p_i

Currency?

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$$

is different (under local unitaries) from

$$|\Psi\rangle = \frac{1}{2}|0\rangle \otimes |0\rangle + \frac{\sqrt{3}}{2}|1\rangle \otimes |1\rangle)$$

If we try to use $|\Psi\rangle$ to teleport a qubit, we can't perfectly teleport (we fail with some probability.)

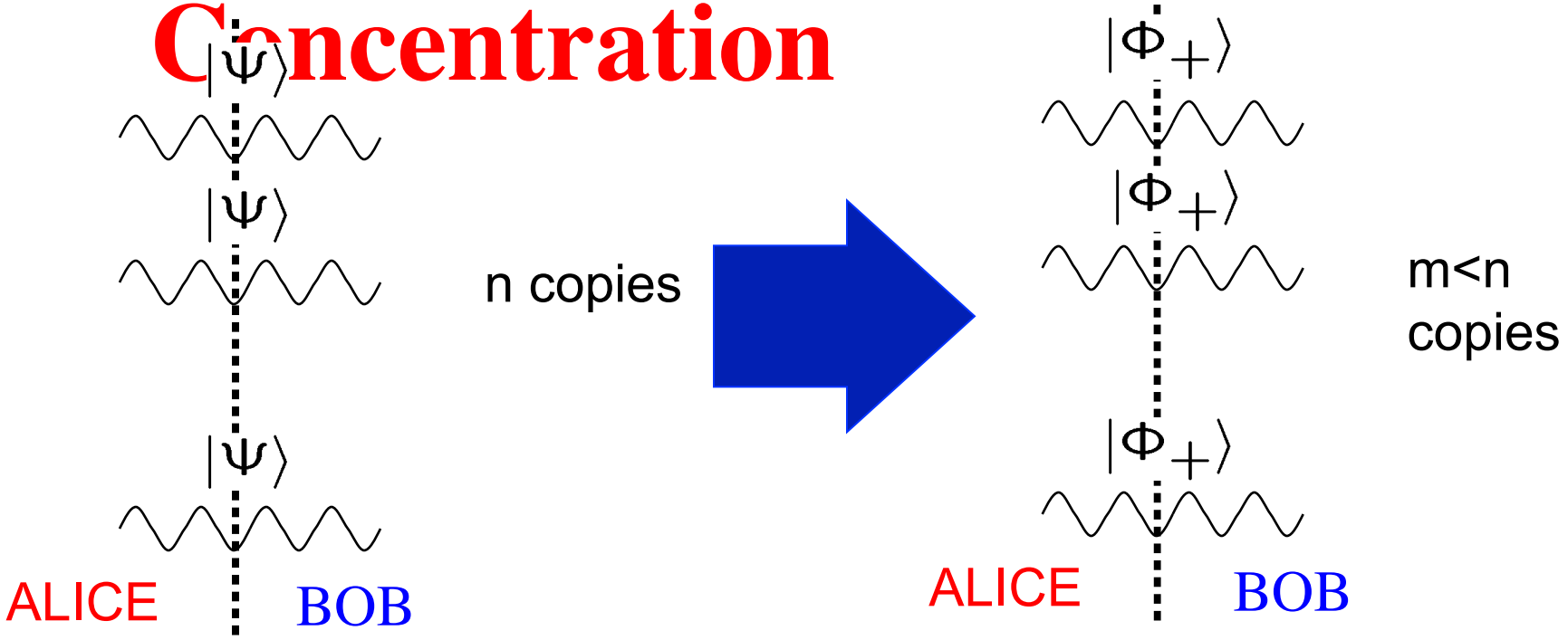
Is $|\Psi\rangle$ equal to some fraction of $|\Phi_+\rangle$?

How many  s equal how many  s?

$|\Psi\rangle$s

$|\Phi_+\rangle$ s

Entanglement Concentration



$$|\Psi\rangle = \sqrt{p}|0\rangle \otimes |0\rangle + \sqrt{1-p}|1\rangle \otimes |1\rangle$$

How to turn

$$|\Psi\rangle^{\otimes n} = \sum_{i=0}^{2^n-1} \sqrt{p}^{|i|}\sqrt{1-p}^{n-|i|}|i\rangle \otimes |i\rangle$$

$|i|$ = number of 0's in bitstring i

into

$$|\Phi_+\rangle^{\otimes m} = \frac{1}{\sqrt{2^m}} \sum_{i=0}^{2^m} |i\rangle \otimes |i\rangle$$

Entanglement Concentration

Measurement's don't have to fully collapse a quantum state:

$|00\rangle$ measurement outcome 1

$|01\rangle$
$|10\rangle$ measurement outcome 2

$|11\rangle$ measurement outcome 3

$$|\Psi\rangle^{\otimes n} = \sum_{i=0}^{2^n-1} \sqrt{p}^{|i|} \sqrt{1-p}^{n-|i|} |i\rangle \otimes |i\rangle$$

Alice measures the number of 0s in bitstring. If outcome is k 0s, state is

$$|\Psi_k\rangle = \frac{1}{\sqrt{\binom{n}{k}}} \sum_{i=0||i|=k}^{2^n-1} |i\rangle \otimes |i\rangle$$

$\log_2 \left[\binom{n}{k}\right]$ ebits!

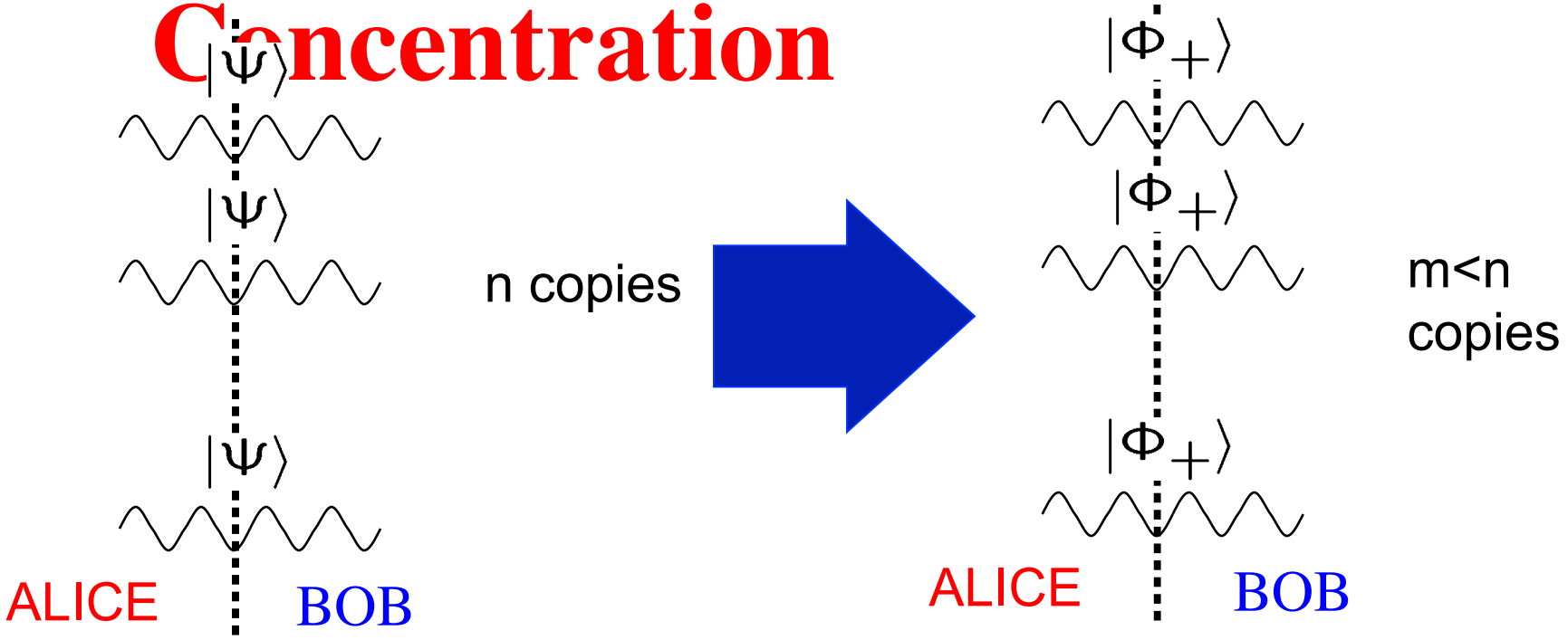
probability of k 0s: $P_k = \binom{n}{k} p^k (1-p)^{n-k}$

Average number of ebits $= \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \log_2 \left[\binom{n}{k}\right]$

$\approx nS(p)$

$$S(p) = -p \log p - (1-p) \log(1-p)$$

Entanglement Concentration



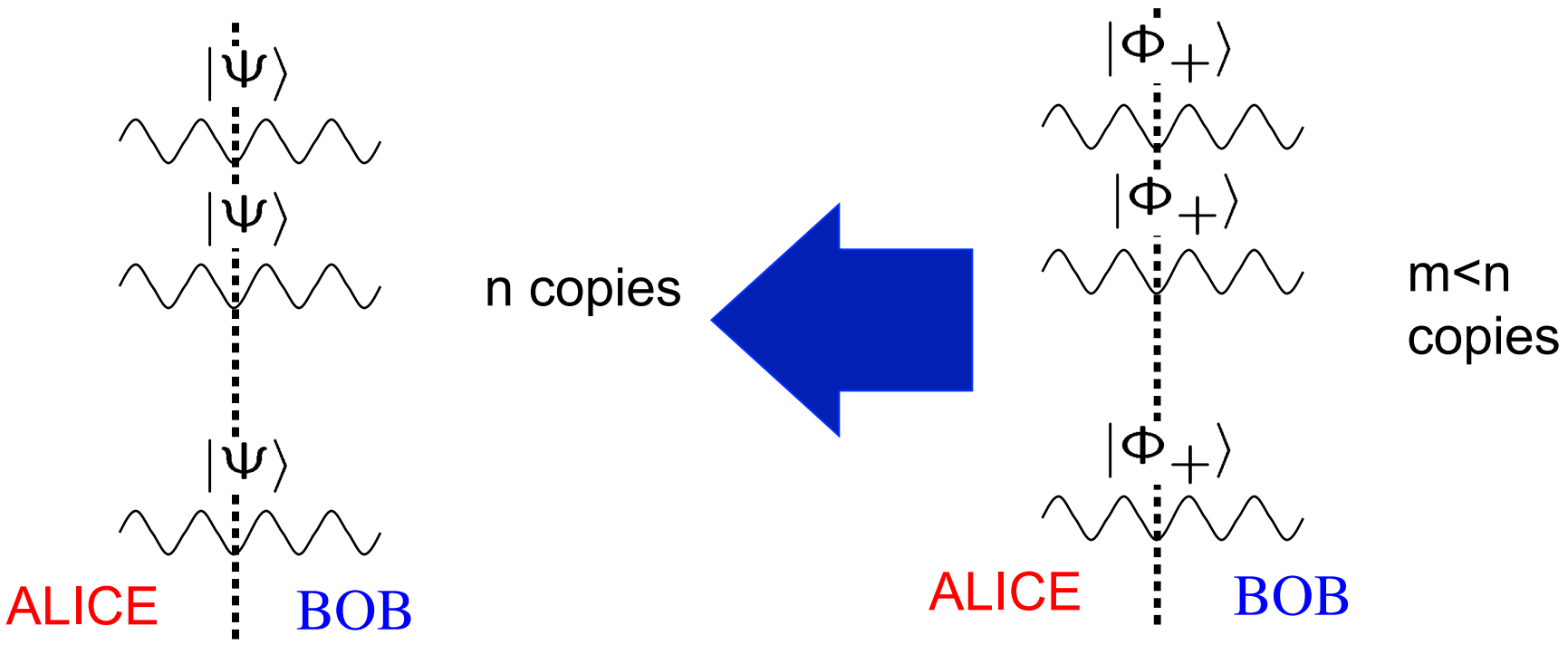
$$|\Psi\rangle = \sqrt{p}|0\rangle \otimes |0\rangle + \sqrt{1-p}|1\rangle \otimes |1\rangle$$

n copies $|\Psi\rangle$ is equal to m copies $|\Phi_+\rangle$

$$m \approx nS(p)$$

entropy of Schmidt coefficients measures entanglement!

Entanglement Dillution



$$|\Psi\rangle = \sqrt{p}|0\rangle \otimes |0\rangle + \sqrt{1-p}|1\rangle \otimes |1\rangle$$

What about the other direction?

How many ebits does it take to produce copies of $|\Psi\rangle$?

Schumacher Compression



Ben Schumacher

One parties version of an entangle pure state

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

is not a pure state, it is mixed:

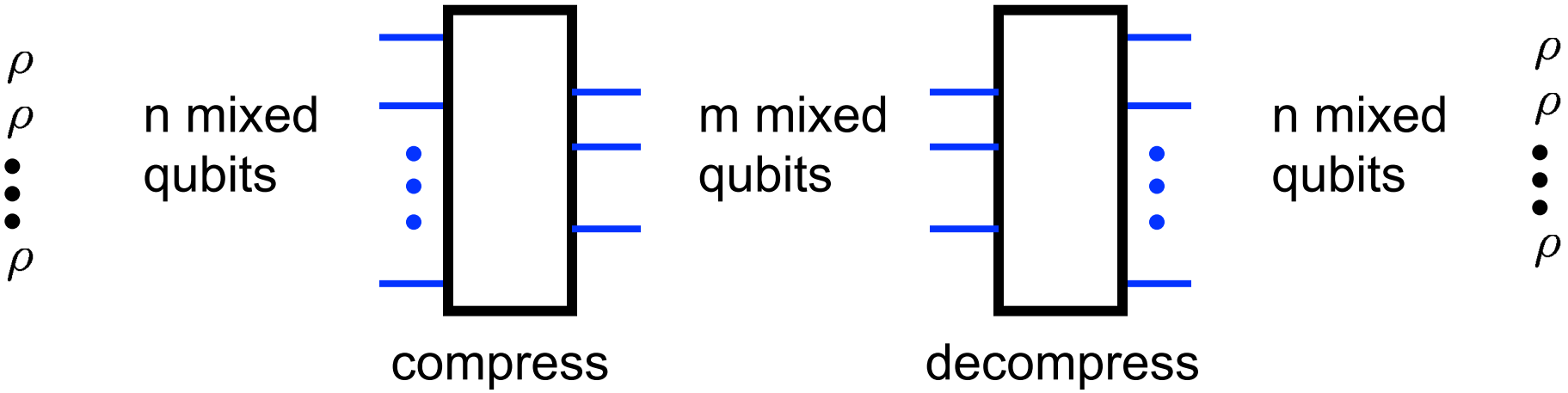
ALICE locally sees her state as

$50\%\ |0\rangle$

$50\%\ |1\rangle$

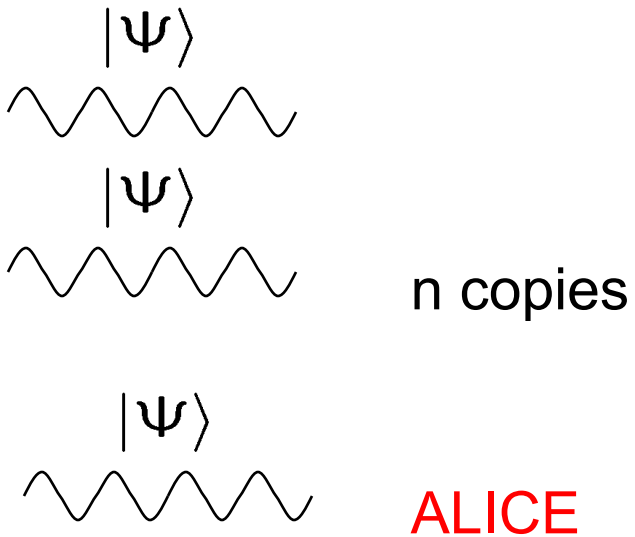
Mixed state: with probability p_i we have one of the orthogonal states $|i\rangle$

Schumacher asked: Can we compress n mixed qubits into a smaller number of mixed qubits?



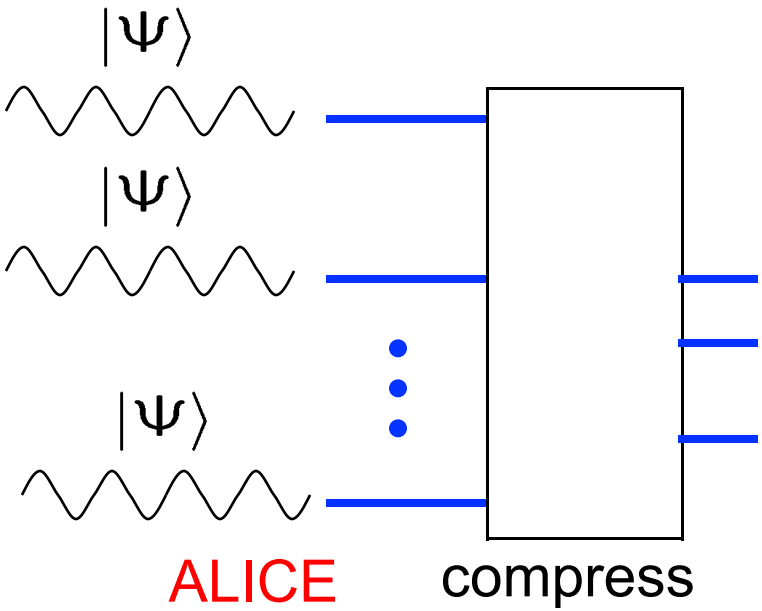
$$m \approx nS(p)$$

Entanglement Dillution



1. Alice locally prepares n copies of

$$|\Psi\rangle = \sqrt{p}|0\rangle \otimes |0\rangle + \sqrt{1-p}|1\rangle \otimes |1\rangle$$

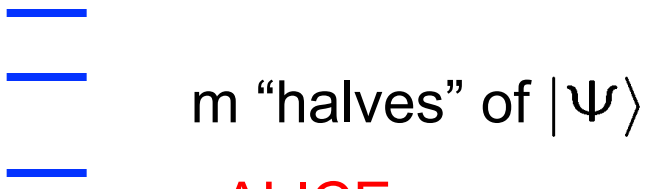


2. Alice takes "Bob's" half and Schumacher compresses the mixed state qubits.

m "halves" of $|\Psi\rangle$

$$m \approx nS(p)$$

Entanglement Dillution



m "halves" of $|\Psi\rangle$

ALICE

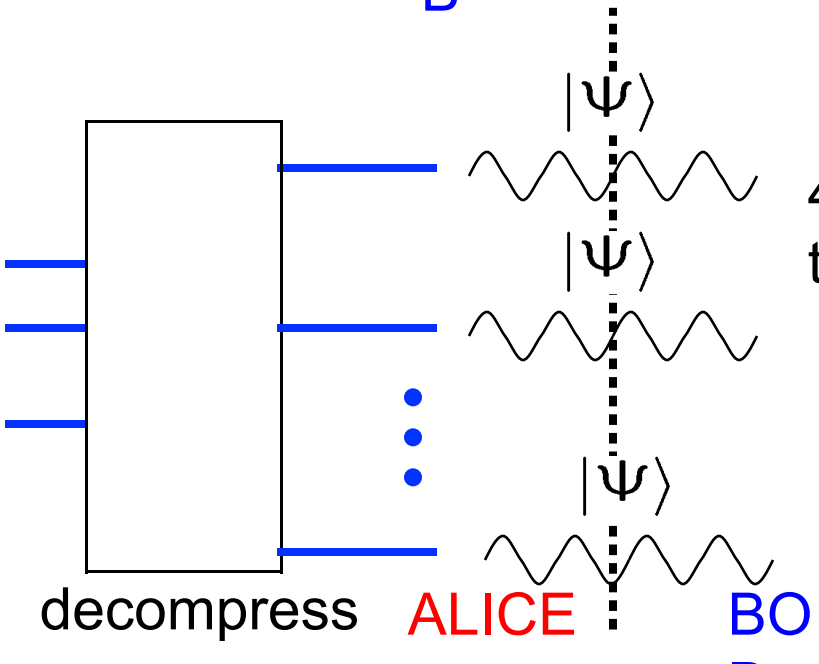
teleportatio
n

3. Alice sends m "halves" of $|\Psi\rangle$ using teleportation. This consumes m ebits.

m "halves" of $|\Psi\rangle$

BO
B

m qubit
"halves" of $|\Psi\rangle$



4. Bob decompresses the "halves"

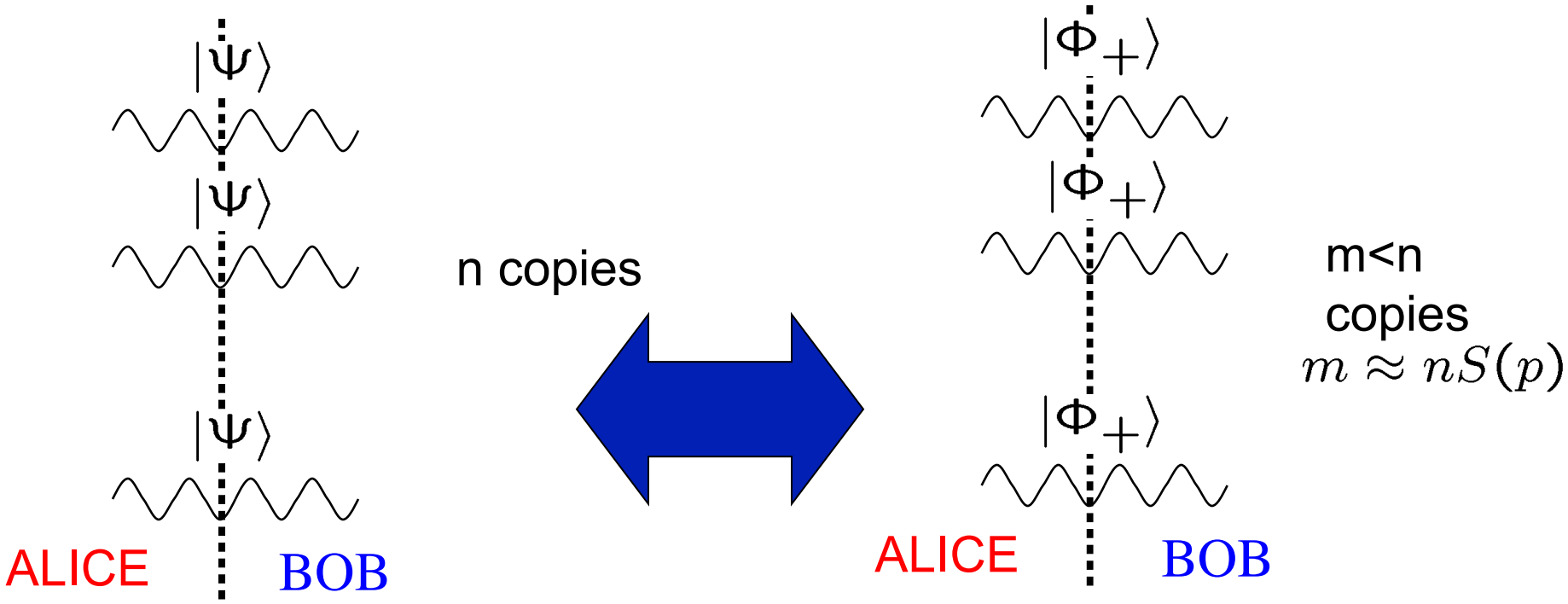
decompress

ALICE

BO

$m \approx nS(p)$
ebits
consumed

Bipartite Pure State Entanglement



$$|\Psi\rangle = \sqrt{p}|0\rangle \otimes |0\rangle + \sqrt{1-p}|1\rangle \otimes |1\rangle$$

$|\Phi_+\rangle$ is the basic currency of two party pure state entanglement

Mixed State Entanglement

not given $|\Psi\rangle$ but $|\Psi_k\rangle$ with probability p_k

Similar theory for converting to and from $|\Phi_+\rangle$?

Entanglement cost of $\rho = \dfrac{\text{ebits consumed to create n copies of } \rho}{\text{n}}$

Entanglement of dilution $\rho = \dfrac{\text{ebits produced from n copies of } \rho}{\text{n}}$

Irreversibility in asymptotic manipulations of entanglement

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Institut für Theoretische Physik, Universität Innsbruck, A-6020 Innsbruck, Austria
(March 28, 2004)

Entanglement of dilution ≤ Entanglement cost

Irreversibility!

Communication Complexity



ALICE

x

F(x,y)

How much (how many bits) do they need to communicate to compute F?



BOB

y

Example

F(x,y)=1 if x=y

F(x,y)=0 otherwise

Alice needs to communicate n bits to Bob

Example

F(x,y)=x⊕y

x=010111

y=10010

1 110010 ← **Odd # 1's F=1**

Even # 1's F=0

Alice needs to communicate 1 bit to Bob

Quantum Communication Complexity



ALICE

How much (how many qubits) do they need to communicate to compute F?

x



y BOB



Ran Raz (1999)

Exponentially less quantum communication than classical communication!

Because of teleportation this implies exponentially less classical communication if we share entanglement!

Fin

Qubits, Qubits, everywhere and not a drop to drink!

1. Entanglement is different from classical correlation.

2. Entanglement is a resource which can be used
   to teleport qubits
   to double the capacity of a qubit channel

3. An ebit is the standard unit of entanglement
   pure state concentration and dilution
   irreversible mixed state cases!

4. Entanglement can be used to cut communication complexity Exponentially!

Dave Bacon, 156 Jorgensen



